CO214 Computer Architecture and Organization Test 2, Spring 2025

Assignment 2,3

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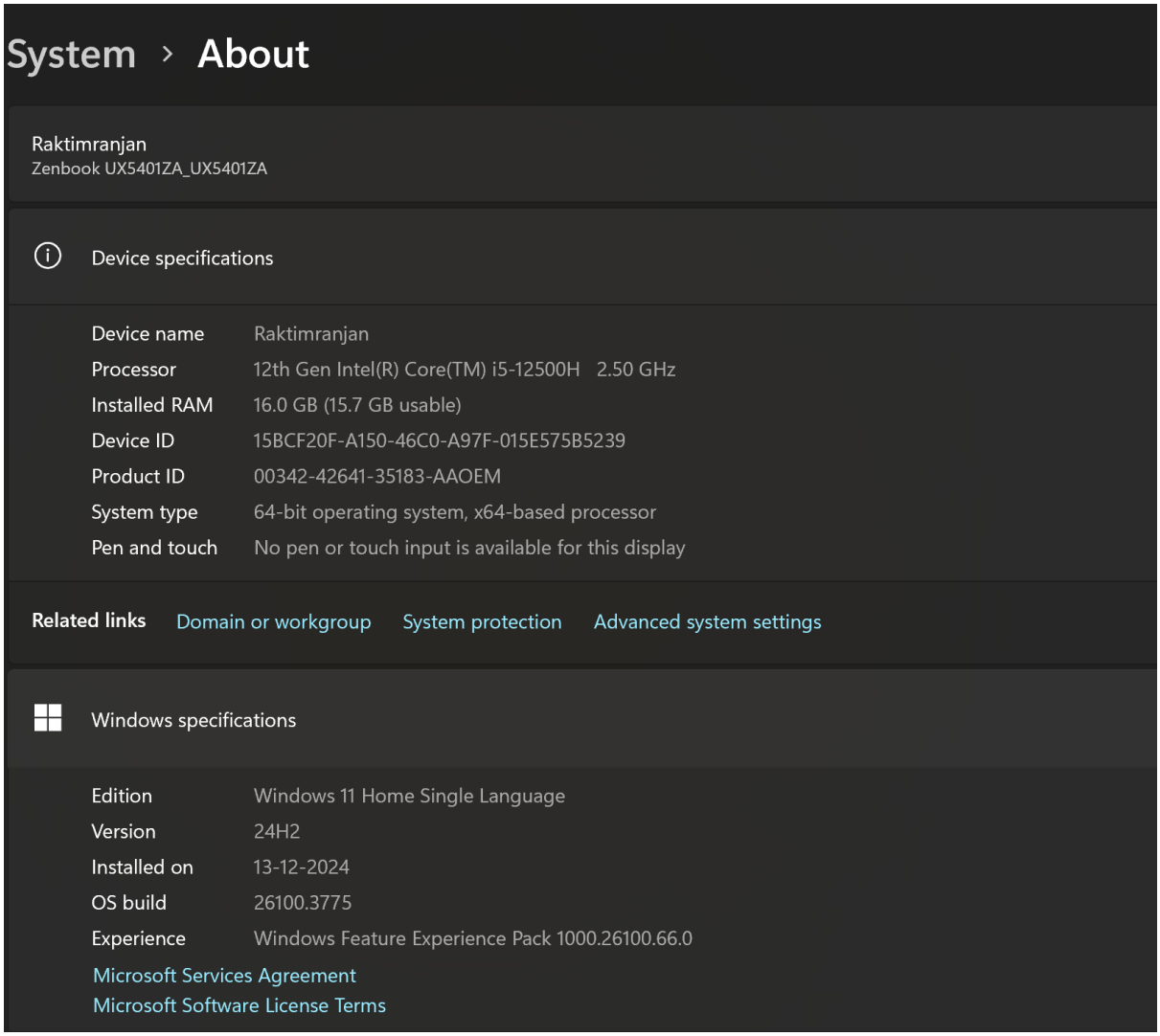
Roll number: CSB23017

Model type: Laptop

Model no: ASUS ZenBook 14x OLED UX5401ZA



Ans 2 :

Hardware Component Specifications :

● **Processor:**

12th Gen Intel Core i5-12500H, 12 cores (4 performance + 8 efficiency), 16 threads, base frequency 2.5GHz (turbo up to 4.5GHz), 18MB Intel Smart Cache, integrated Intel Iris Xe Graphics. Built on Intel 7 process technology with support for DDR5/LPDDR5 memory and PCIe 4.0 lanes for faster data transfer.

● **Memory modules:**

- RAM: 16GB LPDDR5 onboard memory (15.7 GB usable), dual channel, 5200MHz. High bandwidth memory for multitasking and efficient power use.
- Internal Storage: 512GB PCIe Gen4 NVMe M.2 SSD, offers fast read/write speeds and reduced boot times.
- Cache Memory: L1 (80KB per core), L2 (1.25MB per P-core, 2MB per E-core cluster), L3 (shared 18MB Smart Cache).

● **Bus:**

- System Bus: DMI 4.0 (Direct Media Interface) connecting CPU and PCH for high-speed communication.
- Memory Bus: Dual-channel LPDDR5 memory interface providing high data throughput.
- PCIe Bus: PCIe 4.0 x4 lanes for SSD and expansion cards, allowing fast data access and transfer rates.

● **I/O devices:**

- Keyboard: Backlit chiclet keyboard with integrated NumberPad and multi-gesture support touchpad.
- Display: 14.5” 2.8K OLED touchscreen, 120Hz refresh rate, 100% DCI-P3 color gamut, Pantone Validated. Excellent for color-accurate tasks. (Note: No pen or touch input support.)
- Camera: FHD 1080p IR camera with Windows Hello facial recognition.
- Audio: Harman/Kardon certified stereo speakers with Dolby Atmos support, dual-array microphones with AI noise cancellation.
- Cooling: Dual-fan IceCool thermal technology with heat pipes and dynamic performance modes.

● **I/O ports:**

- 2 x Thunderbolt 4 (USB-C) ports: Up to 40Gbps data transfer, Power Delivery 3.0, DisplayPort 1.4
- 1 x USB 3.2 Gen 2 Type-A: 10Gbps high-speed peripheral connectivity
- 1 x HDMI 2.0b: Supports 4K external displays at 60Hz
- 1 x 3.5mm combo audio jack: For microphone and headphones
- 1 x DC-in: For power adapter

Ans 3:

Patent details :

Component type	Component name	Patent no and date	Owned by
Memory	LPDDR5	US20200261169A1, Aug 2020	SK Hynix / JEDEC
IO	Thunderbolt 4	US7779173B2, Aug 2010	Intel Corporation
Bus	PCIe 4.0	US8234536B2, Jul 2012	PCI-SIG
Processor	Intel Core i5-12500H	US10698546B2, Jun 2020	Intel Corporation
Display	OLED Display Tech	US9519728B2, Dec 2016	Samsung Display Co., Ltd
SSD Interface	NVMe (Non-Volatile Memory Express)	US8713163B2, Apr 2014	NVM Express Inc.